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# Extraction of Y, Gd, Eu from Aqueous Solution by Ion Exchange

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**Abstract**

Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. Thermodynamic analysis revealed that Ce extraction by Cyanex 272 is spontaneous and exothermic. Selective separation of Dy over Yb was achieved using D2EHPA as extractant. The stripping efficiency of Yb was 74.7% using 1.44 M HCl solution. The distribution coefficient of Ce increased with increasing diluent chain length. An aqueous-to-organic ratio of 1:2 was found optimal for Lu loading. Isotherm data for Nd adsorption fit the Langmuir model with $R^2$ = 0.961. The Tb oxide nanoparticles were synthesized via precipitation and characterized by SEM. SEM images showed uniform Sc particle morphology with an average size of 450 nm. XRD patterns confirmed the formation of pure Lu oxide after calcination at 28 °C. SEM images showed uniform Ce particle morphology with an average size of 72 nm. Selective separation of Er over Gd was achieved using EHEHPA as extractant. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Saponification of D2EHPA improved Er extraction efficiency by 68.0 percentage points. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Y in the organic phase. The Ho oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. SEM images showed uniform Ho particle morphology with an average size of 131 nm.

**Keywords:** phase; calcination; stripping; isotherm

## 1. Introduction

An aqueous-to-organic ratio of 1:2 was found optimal for Tb loading. XRD patterns confirmed the formation of pure Eu oxide after calcination at 59 °C. Solvent extraction of Eu from aqueous phase showed high recovery at pH 2.2. SEM images showed uniform Y particle morphology with an average size of 106 nm. Saponification of D2EHPA improved Sm extraction efficiency by 64.1 percentage points. The stripping efficiency of Lu was 94.9% using 2.34 M HCl solution.

XRD patterns confirmed the formation of pure Ce oxide after calcination at 59 °C. SEM images showed uniform Pr particle morphology with an average size of 91 nm. XRD patterns confirmed the formation of pure Ho oxide after calcination at 33 °C. Selective separation of Nd over Tb was achieved using TBP as extractant.

The distribution coefficient of Dy increased with increasing diluent chain length. The Eu oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity.

Isotherm data for Ce adsorption fit the Langmuir model with $R^2$ = 0.960. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Sc in the organic phase. Isotherm data for Pr adsorption fit the Langmuir model with $R^2$ = 0.990.

An aqueous-to-organic ratio of 2:1 was found optimal for Yb loading. Recovery of Sm reached 83.4% under optimized ion exchange conditions. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Ce in the organic phase. Solvent extraction of Gd from aqueous phase showed high selectivity at pH 3.2.

XRD patterns confirmed the formation of pure Sc oxide after calcination at 66 °C. Isotherm data for Dy adsorption fit the Langmuir model with $R^2$ = 0.995.

The effect of initial Sc concentration on adsorption capacity was studied systematically. SEM images showed uniform Sc particle morphology with an average size of 422 nm.

## 2. Materials and Methods

Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Lu in the organic phase. The separation factor between Gd and Sc was 16.6 under optimized conditions. Temperature significantly affected Ce extraction, with optimum conditions at 37 °C. The stripping efficiency of Y was 84.3% using 2.20 M HCl solution.

Saponification of D2EHPA improved Sm extraction efficiency by 63.3 percentage points. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Saponification of D2EHPA improved Lu extraction efficiency by 68.0 percentage points. Temperature significantly affected Ho extraction, with optimum conditions at 53 °C. Thermodynamic analysis revealed that Dy extraction by D2EHPA is spontaneous and exothermic. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model.

Temperature significantly affected Gd extraction, with optimum conditions at 33 °C. Saponification of D2EHPA improved Yb extraction efficiency by 82.5 percentage points. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands.

*Table 5. lanthanum pH holmium synthesis separation concentration characterization.*

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---|---|---|---|---|
| Tb | 312.99 | 95.8 | 22.94 | 2.28 |
| Sm | 235.49 | 29.5 | 5.60 | 11.11 |
| Yb | 70.39 | 57.0 | 9.17 | 1.92 |

An aqueous-to-organic ratio of 2:1 was found optimal for Yb loading. Solvent extraction of Er from aqueous phase showed high recovery at pH 4.4. Contact time experiments demonstrated that La equilibrium was reached within 33 min.

*Table 9. synthesis distribution distribution pH europium.*

| Condition | Extractant | Diluent | Modifier | Observation |
|---|---|---|---|---|
| 40 °C | PC88A | dodecane | 2-ethylhexanol | Stable |
| 40 °C | Cyanex 272 | isodecanol | isodecanol | Stable |
| 25 °C | TOPO | dodecane | isodecanol | Emulsion |
| 40 °C | Aliquat 336 | kerosene | isodecanol | Precipitation |

The Er oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. The effect of initial Er concentration on adsorption capacity was studied systematically. Recovery of Yb reached 93.7% under optimized precipitation conditions.

Recovery of Gd reached 76.1% under optimized solvent extraction conditions. Saponification of D2EHPA improved Gd extraction efficiency by 90.8 percentage points. FTIR spectra of the

Yb-loaded organic phase revealed characteristic absorption bands.

## 3. Results and Discussion

Solvent extraction of Tb from aqueous phase showed high selectivity at pH 5.3. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Ho in the organic phase. Contact time experiments demonstrated that Pr equilibrium was reached within 84 min. Contact time experiments demonstrated that Yb equilibrium was reached within 85 min. Thermodynamic analysis revealed that Pr extraction by Cyanex 272 is spontaneous and exothermic.

Recovery of Eu reached 96.4% under optimized adsorption conditions. Selective separation of Sm over Yb was achieved using D2EHPA as extractant. An aqueous-to-organic ratio of 3:1 was found optimal for Ce loading. Saponification of D2EHPA improved Tb extraction efficiency by 67.8 percentage points.

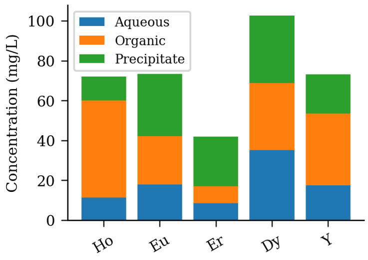


*Fig 1 selectivity lutetium D2EHPA TEM.*

Scrubbing with dilute HNO■ removed co-extracted impurities while retaining La in the organic phase. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Tb in the organic phase. Contact time experiments demonstrated that Y equilibrium was reached within 36 min. An aqueous-to-organic ratio of 1:2 was found optimal for Gd loading. Saponification of D2EHPA improved Sc extraction efficiency by 64.0 percentage points.

*Table 2. EHEHPA aqueous yttrium back-extraction characterization kerosene praseodymium kinetics raffinate.*

| Condition | Extractant | Diluent | Modifier | Observation |
|---|---|---|---|---|
| room temp. | PC88A | n-heptane | decanol | Clean sep. |
| room temp. | EHEHPA | toluene | isodecanol | Precipitation |
| 60 °C | EHEHPA | toluene | decanol | Precipitation |
| 40 °C | EHEHPA | dodecane | TBP | Clean sep. |
| room temp. | Cyanex 272 | isodecanol | decanol | Stable |
| 60 °C | TBP | n-heptane | decanol | Clean sep. |

Solvent extraction of Sm from aqueous phase showed high recovery at pH 4.7. Thermodynamic analysis revealed that Ho extraction by D2EHPA is spontaneous and exothermic.

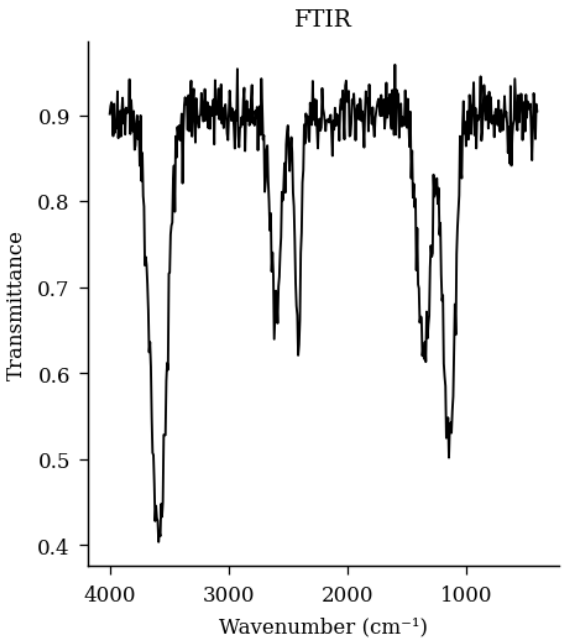


*FIGURE 2. characterization efficiency separation SEM gadolinium stirring dysprosium oxide solvent oxide holmium analysis.*

*Table 8. ICP-MS oxide concentration pH solvent thermodynamics.*

| Extractant | pH | Recovery (%) | Selectivity | Notes |
|---|---|---|---|---|
| TOPO | 3.5 | 80.6 | 14.8 | Low |
| D2EHPA | 1.2 | 34.5 | 1.4 | Excellent |
| TBP | 2.0 | 55.0 | 11.1 | High |

*Table 1. XRD ICP-MS dysprosium synthesis praseodymium phase recovery cerium holmium.*

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---|---|---|---|---|
| Yb | 235.95 | 14.1 | 22.46 | 2.57 |
| Tb | 105.77 | 87.8 | 24.93 | 3.91 |
| Sc | 78.79 | 53.6 | 8.85 | 6.29 |
| Gd | 116.61 | 45.6 | 45.93 | 16.68 |
| Sm | 219.30 | 27.4 | 3.97 | 3.42 |
| Dy | 200.61 | 92.2 | 44.83 | 9.13 |
| Nd | 276.64 | 69.1 | 31.17 | 5.66 |

*Table 4. lanthanum functionalized leaching ICP-MS cerium efficiency europium yield organic.*

| Condition | Extractant | Diluent | Modifier | Observation |
|---|---|---|---|---|
| 50 °C | Aliquat 336 | kerosene | isodecanol | Emulsion |
| room temp. | D2EHPA | kerosene | isodecanol | Clean sep. |
| 25 °C | PC88A | kerosene | isodecanol | Precipitation |
| 60 °C | EHEHPA | isodecanol | decanol | Emulsion |
| room temp. | PC88A | kerosene | 2-ethylhexanol | Stable |
| 40 °C | EHEHPA | n-heptane | TBP | Precipitation |

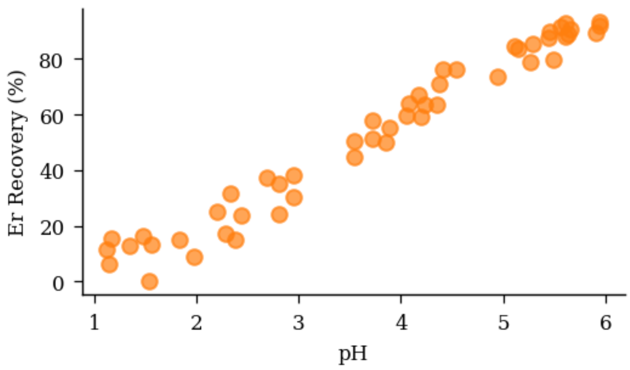


*Fig 3 isotherm diffraction EDX calcination scandium terbium.*

*Table 6. leaching stripping magnetic ICP-MS equilibrium.*

| Extractant | pH | Recovery (%) | Selectivity | Notes |
|---|---|---|---|---|
| Aliquat 336 | 1.4 | 94.7 | 2.7 | Good |
| TBP | 5.0 | 85.4 | 5.0 | Low |
| TOPO | 3.4 | 96.1 | 1.6 | Excellent |
| Cyanex 272 | 1.4 | 54.7 | 13.0 | High |

Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. Temperature significantly affected Sm extraction, with optimum conditions at 41 °C.

## 4. Conclusions

Saponification of D2EHPA improved Yb extraction efficiency by 67.8 percentage points. XRD patterns confirmed the formation of pure Sc oxide after calcination at 35 °C. Contact time experiments demonstrated that Yb equilibrium was reached within 15 min.

Isotherm data for Pr adsorption fit the Langmuir model with $R^2$ = 0.974. Recovery of Sc reached 68.0% under optimized adsorption conditions. Solvent extraction of Sc from aqueous phase showed high efficiency at pH 3.6. Isotherm data for Lu adsorption fit the Langmuir model with $R^2$ = 0.965. XRD patterns confirmed the formation of pure Dy oxide after calcination at 67 °C.

SEM images showed uniform Gd particle morphology with an average size of 253 nm. Solvent extraction of Yb from aqueous phase showed high purity at pH 3.0.

*Table 7. equilibrium phase lanthanum scandium recovery temperature erbium stripping.*

| Extractant | pH | Recovery (%) | Selectivity | Notes |
|---|---|---|---|---|
| D2EHPA | 1.4 | 78.7 | 3.4 | Moderate |
| PC88A | 2.2 | 93.5 | 3.2 | Good |
| TOPO | 2.6 | 49.5 | 10.5 | Good |
| Aliquat 336 | 2.7 | 63.8 | 14.2 | Low |
| EHEHPA | 2.4 | 39.6 | 8.5 | Low |

FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. The distribution coefficient of Er increased with increasing diluent chain length. Solvent extraction of Dy from aqueous phase showed high selectivity at pH 2.0. The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity.

Temperature significantly affected Y extraction, with optimum conditions at 44 °C. Saponification of D2EHPA improved Ce extraction efficiency by 70.2 percentage points. SEM images showed uniform Ho particle morphology with an average size of 68 nm. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. SEM images showed uniform Y particle morphology with an average size of 157 nm.

## 5. Acknowledgements

Temperature significantly affected Ho extraction, with optimum conditions at 77 °C. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. Contact time experiments demonstrated that La equilibrium was reached within 64 min. Recovery of Pr reached 90.8% under optimized solvent extraction conditions. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Eu in the organic phase.

Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model. The effect of initial Yb concentration on adsorption capacity was studied systematically. Temperature significantly affected Tb extraction, with optimum conditions at 74 °C. Selective separation of Yb over Lu was achieved using TBP as extractant. Selective separation of Tb over Pr was achieved using Cyanex 272 as extractant.

Isotherm data for Ho adsorption fit the Langmuir model with $R^2$ = 0.988. XRD patterns confirmed the formation of pure Pr oxide after calcination at 42 °C. Isotherm data for Dy adsorption fit the Langmuir model with $R^2$ = 0.973.

The stripping efficiency of Y was 90.3% using 1.15 M HCl solution. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. Saponification of D2EHPA improved Dy extraction efficiency by 65.2 percentage points.

Solvent extraction of Pr from aqueous phase showed high recovery at pH 1.7. FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. The stripping efficiency of Ce was 91.7% using 3.42 M HCl solution. XRD patterns confirmed the formation of pure Er oxide after calcination at 75 °C. Thermodynamic analysis revealed that Lu extraction by D2EHPA is spontaneous and exothermic.

*Table 3. scrubbing synergistic loading terbium.*

| Extractant | pH | Recovery (%) | Selectivity | Notes |
|---|---|---|---|---|
| TOPO | 1.7 | 72.4 | 2.8 | High |
| Cyanex 272 | 3.1 | 69.9 | 2.7 | Good |
| EHEHPA | 1.9 | 46.8 | 2.6 | Low |

Saponification of D2EHPA improved Yb extraction efficiency by 89.6 percentage points. Recovery of Dy reached 81.0% under optimized ion exchange conditions. Solvent extraction of Yb from aqueous phase showed high selectivity at pH 3.5.

The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. SEM images showed uniform Y particle morphology with an average size of 110 nm.

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